Lightweight Q-Learning Based Joint RACH Optimization for Ultra-Dense NB-IoT Smart Meter Networks

***Abstract*—Demand-response events in NB-IoT-enabled Advanced Metering Infrastructure (AMI) can trigger simultaneous uplink access attempts from tens of thousands of smart meters, causing severe Random-Access Channel (RACH) congestion that static Third Generation Partnership Project (3GPP) configurations cannot efficiently manage. Existing reinforcement learning (RL)-based approaches typically optimise a single RACH parameter and often fail to maximise successful network access under highly dynamic traffic conditions. This paper proposes a lightweight tabular Q-learning framework that jointly optimises three 3GPP-configurable RACH parameters: the Access Class Barring (ACB) factor, the Backoff Indicator (BI), and the NPRACH preamble group configuration. The optimisation problem is formulated as a 150-state, 120-action Markov Decision Process (MDP) using a throughput-oriented reward function and a queue-aware delay model to improve access performance while mitigating reward-hacking behaviour. Simulation results for an ultra-dense deployment comprising $N_{UE} = 50K$ smart meters demonstrate an access success probability of $Ps = 0.592$, an average access delay of 1,534 ms, and zero service-level agreement (SLA) violations under a 30 s latency constraint. The proposed controller requires only a 72 KB Q-table and deterministic $O(1)$ execution, making it suitable for real-time deployment within commercial NB-IoT eNodeB schedulers.**

Keywords*—Narrowband Internet of Things (NB-IoT); Massive Machine-Type Communications (mMTC); Random Access Channel (RACH); Q-Learning; Advanced Metering Infrastructure (AMI); RACH Congestion Control.

I. INTRODUCTION

The rapid digitalisation of electric power systems has accelerated the deployment of AMI, enabling automated meter reading, real-time energy monitoring, demand-response (DR), outage management, and intelligent grid operation. Supporting millions of geographically distributed smart meters requires communication networks that provide ubiquitous coverage, long battery lifetime, high reliability, and massive connectivity. Among Low-Power Wide-Area Network (LPWAN) technologies, Narrowband Internet of Things (NB-IoT) has become a leading cellular solution for smart metering owing to its extended coverage, low power consumption, and seamless integration with existing Long-Term Evolution (LTE) infrastructure [1], [2], [3]. Consequently, NB-IoT-enabled smart meters are increasingly deployed for periodic energy reporting, remote device management, and event-driven services defined by the IEC 62056 DLMS/COSEM framework [4], [5]. Although [5] periodic smart meter reporting generates predictable traffic, DR events, outage restoration, and large-scale firmware updates can trigger highly synchronised uplink access attempts from tens of thousands of devices within a short interval. These traffic bursts create severe RACH congestion, resulting in excessive NPRACH preamble collisions, repeated retransmissions, increased access delay, and degraded network throughput [6], [7]. The 3GPP random access procedure provides configurable congestion-control mechanisms, including the Access Class Barring (ACB) factor, Backoff Indicator (BI), and NPRACH resource configuration. However, these parameters are typically configured using static or semi-static policies that cannot adequately adapt to rapidly varying traffic conditions in ultra-dense NB-IoT smart metering deployments.

Recent studies have investigated analytical modelling, adaptive congestion control, and reinforcement learning (RL) to improve NB-IoT RACH performance [7], [8], [9], [10]. Although RL enables adaptive access control without requiring explicit traffic models, existing approaches generally optimise only a single 3GPP-configurable RACH parameter, rely on computationally intensive deep RL architectures, or overlook practical deployment constraints such as deterministic execution, limited memory resources, and realistic queue evolution [7], [9], [10], [11], [12]. Consequently, a lightweight framework capable of jointly optimising multiple RACH control parameters while remaining suitable for real-time deployment in commercial radio access networks is still lacking.

Motivated by these challenges, this paper proposes a lightweight tabular Q-learning framework that jointly optimises the Access Class Barring (ACB) factor, Backoff Indicator (BI), and NPRACH preamble group configuration. The optimisation problem is formulated as a compact Markov Decision Process (MDP) using a throughput-oriented reward function and a three-component delay model based on a pseudo-stationary application of Little's Law. The resulting controller achieves deterministic execution with minimal computational and memory overhead, enabling practical deployment within commercial NB-IoT eNodeB scheduling and radio resource management (RAN) functions.

The principal contributions of this paper are summarised as follows.

- To the best of our knowledge, this is the first study to jointly optimise the Access Class Barring factor,

Backoff Indicator, and NPRACH preamble group configuration using a lightweight tabular Q-learning framework for an ultra-dense NB-IoT-enabled AMI network comprising $N_{\mathrm{UE}} = 50{,}000$ smart meters.

- A three-component delay model based on a pseudo-stationary application of Little's Law is developed to provide a realistic representation of large-scale RACH congestion while preserving queue evolution.
- A throughput-oriented reward formulation and mass-conserving queue model are proposed to mitigate reward-hacking behaviour, improve access success probability, and ensure that deferred and collided requests remain available for subsequent contention.
- The proposed framework requires only a 72 KB Q-table and deterministic $O(1)$ execution, enabling real-time optimisation of 3GPP-configurable RACH parameters within commercial NB-IoT eNodeB schedulers.

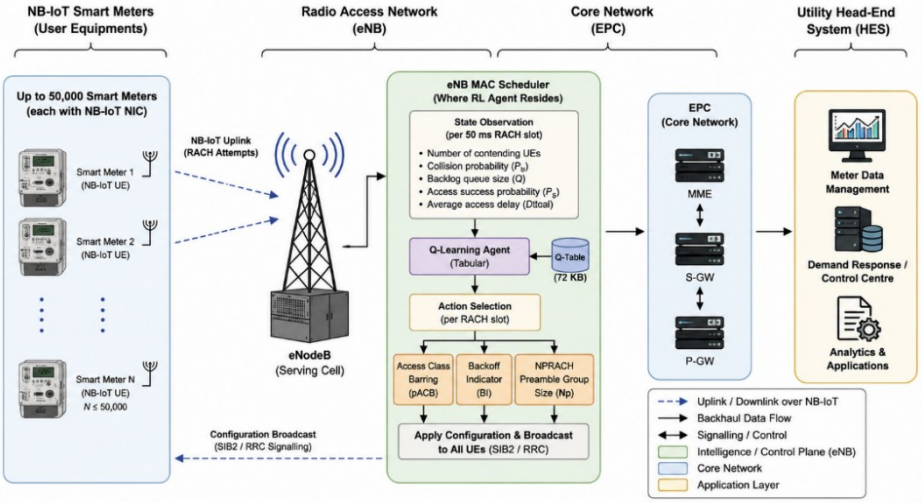


Figure 1: Proposed NB-IoT-enabled AMI communication architecture and network-side reinforcement learning framework.

II. Related Work and Research Gap

Efficient RACH congestion control has become a key research topic in Narrowband Internet of Things (NB-IoT) networks owing to the rapid growth of massive machine-type communications (mMTC) and large-scale smart metering deployments. Early research primarily focused on analytical modelling of collision probability, access delay, and network scalability, establishing the theoretical foundations for congestion-control mechanisms such as Access Class Barring (ACB), adaptive backoff, and dynamic NPRACH resource configuration [7], [13]. These studies significantly improved the understanding of random access behaviour but generally relied on static or semi-static parameter settings that are unable to respond effectively to rapidly changing traffic conditions.

More recent investigations have concentrated on NB-IoT-enabled smart metering, evaluating coverage enhancement techniques, large-scale Advanced Metering Infrastructure (AMI) deployments, and practical radio propagation conditions [7], [9]. These studies demonstrated the feasibility of NB-IoT for utility communications and highlighted the influence of network configuration on communication reliability. However, the proposed optimisation strategies typically consider individual RACH parameters independently and provide limited adaptability to the highly synchronised traffic bursts generated during demand-response events, outage restoration, or large-scale firmware updates.

To improve congestion adaptation, recent studies have increasingly explored reinforcement learning (RL) and machine learning (ML) techniques for dynamic RACH optimisation. Existing approaches optimise individual RACH parameters, estimate traffic demand, or perform proactive radio resource allocation to improve access success probability and channel utilisation [10], [12], [14]. Although these methods demonstrate promising performance improvements, several practical challenges remain. Most reported solutions optimise only a single 3GPP-configurable RACH parameter, rely on computationally intensive deep neural networks, or employ simplified queue evolution and collision-oriented reward formulations that may encourage excessive access barring instead of maximising successful smart meter connectivity [7]. Consequently, a lightweight and deployment-oriented framework capable of jointly optimising multiple 3GPP-configurable RACH parameters while maintaining deterministic execution and low computational overhead remains largely unexplored. To address this research gap, this paper proposes a lightweight tabular Q-learning framework that jointly optimises the Access Class Barring factor, the Backoff Indicator, and the NPRACH preamble group configuration within a unified Markov Decision

Process (MDP), providing a practical solution for ultra-dense NB-IoT smart metering networks.

III. System Model and Problem Formulation

This section presents the NB-IoT smart metering communication model adopted for large-scale AMI networks. It describes the network architecture, traffic model, and mathematical formulation of RACH congestion under periodic and event-driven smart meter communications. These models define the optimisation framework developed in Section IV.

A. NB-IoT Smart Meter Communication Architecture

Error! Reference source not found. Figure 1 illustrates the NB-IoT-enabled Advanced Metering Infrastructure (AMI) communication architecture considered in this work. Each electricity smart meter incorporates an embedded NB-IoT Network Interface Card (NIC) operating as a standards-compliant Third Generation Partnership Project (3GPP) User Equipment (UE). Meter readings, event notifications, and demand-response messages are transmitted over the NB-IoT air interface to the serving evolved Node B (eNB), which forwards the data through the Evolved Packet Core (EPC) to the utility Head-End System (HES). The proposed congestion-control framework is implemented entirely within the Radio Access Network (RAN). During each NPRACH access interval, the eNB scheduler observes the network state, including the number of contending devices, collision level, and backlog, before jointly selecting the Access Class Barring (ACB) factor, the Backoff Indicator (BI), and the NPRACH resource configuration. These parameters are broadcast using standard NB-IoT system information signalling, allowing all smart meters to continue executing the standard 3GPP random access procedure without modification. Consequently, the proposed framework preserves compatibility with commercially deployed NB-IoT smart meters while enabling adaptive network-side RACH optimisation.

B. Network Configuration

A single NB-IoT cell serving $N_{\mathrm{UE}} = 50{,}000$ static smart meters is considered, representing an ultra-dense Advanced Metering Infrastructure (AMI) deployment consistent with the massive machine-type communications (mMTC) device-density targets defined in ETSI TS 122.368. Each smart meter periodically transmits energy consumption data using the IEC 62056 DLMS/COSEM communication framework and may additionally generate event-driven traffic during demand-response operations, outage notifications, or alarm conditions.

The network follows the standard 3GPP NB-IoT random access procedure with a Random Access Channel (RACH) interval of $T_{\mathrm{RACH}} = 50$ ms. During each RACH interval, the evolved Node B (eNB) scheduler jointly configures the Access Class Barring probability p_{ACB}, the Backoff Indicator BI, and the NPRACH preamble group configuration N_{p}, which constitute the action variables of the proposed reinforcement learning framework. The remaining simulation parameters are summarised in Table I.

To evaluate latency performance, a study-specific service-level agreement (SLA) threshold of $D_{\mathrm{SLA}} = 30{,}000$ ms is adopted. The corresponding SLA violation ratio is defined as

$$\Phi = \frac{1}{T_{\mathrm{active}}} \sum_{t=1}^{T_{\mathrm{sim}}} \mathbf{1}\left(D_{\mathrm{total}}(t) > D_{\mathrm{SLA}}\right) \quad (1)$$

Where $D_{\mathrm{total}}(t)$ denotes the total access delay experienced during slot t and $\mathbf{1}(\cdot)$ is the indicator function.

C. Smart Meter Traffic Model

The traffic model comprises periodic smart meter reporting and event-driven demand-response traffic. The total number of newly generated access requests during scheduling interval t is

$$M_{\mathrm{new}}(t) = M_{\mathrm{periodic}}(t) + M_{\mathrm{burst}}(t) \quad (2)$$

where $M_{\mathrm{periodic}}(t)$ represents periodic IEC 62056 DLMS/COSEM meter reporting and $M_{\mathrm{burst}}(t)$ denotes synchronised demand-response traffic. Following the 3GPP system-level evaluation methodology, burst arrivals are modelled using a Beta-distributed temporal profile [10], producing high contention immediately after a broadcast control message. The total number of devices contending for Random Access Channel (RACH) resources is therefore

$$M_{\mathrm{cont}}(t) = M_{\mathrm{new}}(t) + Q_{\mathrm{back}}(t) \quad (3)$$

where $Q_{\mathrm{back}}(t)$ denotes the backlog of devices awaiting retransmission following unsuccessful access attempts.

D. Mathematical Formulation of the NB-IoT RACH Congestion Model

The offered load on the NPRACH) during scheduling interval t is defined as

$$g(t) = \frac{M_{\mathrm{cont}}(t)}{N_{\mathrm{p}}(t)} \quad (4)$$

where $M_{\mathrm{cont}}(t)$ is the number of contending devices and $N_{\mathrm{p}}(t)$ denotes the available NPRACH preamble resources. Larger values of $g(t)$ increase contention, resulting in higher collision probability and additional retransmissions.

$$D_{\mathrm{total}} = D_{\mathrm{RACH}} + D_{\mathrm{queue}} + D_{\mathrm{backoff}} \quad (5)$$

where D_{RACH} represents contention delay during random access, D_{queue} denotes queueing delay estimated using Little's Law, and D_{backoff}corresponds to retransmission delay introduced by the Backoff Indicator. This decomposition captures the dominant delay mechanisms during large-scale NB-IoT RACH congestion while maintaining computational efficiency. To preserve traffic conservation, unsuccessful access attempts remain in the retransmission backlog until successful channel acquisition. Consequently, all generated smart meter traffic is retained throughout the simulation, providing

realistic congestion evolution under ultra-dense NB-IoT deployments.

E. Problem Formulation

The objective is to adaptively configure the 3GPP-controllable Random-Access Channel (RACH) parameters to maximise successful network access while maintaining queue stability under ultra-dense NB-IoT smart metering traffic. During each RACH scheduling interval, the evolved Node B (eNB) jointly configures the Access Class Barring (ACB) factor, the Backoff Indicator (BI), and the NPRACH preamble group configuration. The optimisation action vector is defined as

$$\mathbf{a}(t) = \left[p_{\mathrm{ACB}}(t), BI(t), N_{\mathrm{p}}(t)\right] \quad (6)$$

Where $p_{\mathrm{ACB}}(t)$, $BI(t)$, and $N_{\mathrm{p}}(t)$ denote the selected Access Class Barring factor, Backoff Indicator, and NPRACH preamble group configuration during scheduling interval t. The optimisation objective is formulated as

$$\max_{\pi} \lim_{T\to\infty} \frac{1}{T} \sum_{t=0}^{T-1} S(t) \quad (7)$$

where π denotes the scheduling policy and $S(t)$ represents the number of successful random access procedures during scheduling interval t. The optimisation is performed subject to the operational constraints imposed by the NB-IoT random access procedure, $0 \le p_{\mathrm{ACB}} \le 1, BI \in B, N_{\mathrm{p}} \in \mathcal{N}, D_{\mathrm{total}} \le D_{\mathrm{SLA}}$ and $Q(t) \ge 0, \forall t$, where $\mathcal{B}$ and $\mathcal{N}$ denote the admissible Backoff Indicator and NPRACH configuration sets defined by the 3GPP specifications, D_{SLA} is the service-level delay threshold, and $Q(t)$ is the retransmission backlog. Because the optimisation problem involves stochastic traffic arrivals, retransmissions, and queue evolution, obtaining a closed-form optimal policy is computationally intractable for ultra-dense deployments. Therefore, the optimisation problem is solved using the lightweight tabular Q-learning framework described in Section IV.

IV. Proposed Lightweight Q-Learning Framework

Static or semi-static RACH parameter configuration cannot efficiently accommodate the highly dynamic traffic generated by ultra-dense NB-IoT smart metering networks. To address this limitation, a lightweight tabular Q-learning framework is proposed to jointly optimise the three configurable 3GPP RACH parameters at the evolved Node B (eNB): the Access Class Barring (ACB) factor, the Backoff Indicator (BI), and the Narrowband Physical Random Access Channel (NPRACH) preamble group configuration. During each RACH scheduling interval, the eNB scheduler observes the current network state, selects a joint RACH configuration, receives a reward based on the resulting network performance, and updates the action-value table. Figure 2 illustrates the interaction between the learning agent and the NB-IoT network.

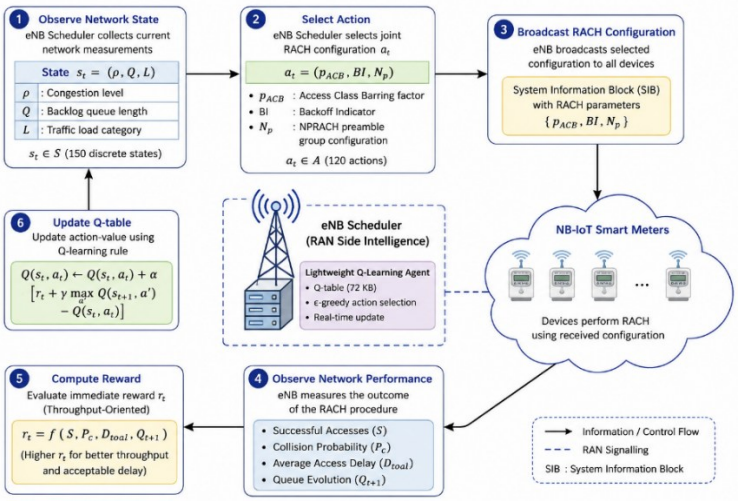


Figure 2: Overall workflow of the proposed lightweight Q-learning based RACH optimisation framework.

A. Markov Decision Process Formulation

The congestion-control problem is formulated as a finite Markov Decision Process (MDP)

$$\mathcal{M} = (\mathcal{S}, \mathcal{A}, \mathcal{P}, \mathcal{R}, \gamma) \quad (8)$$

where S denotes the finite state space, A is the finite action space, P represents the state-transition probability, R is the immediate reward function, and γ denotes the discount factor. At scheduling interval t, the eNB observes the network state $s_t \in \mathcal{S}$, selects an action $a_t \in \mathcal{A}$, and receives the immediate reward $r_t = \mathcal{R}(s_t, a_t)$. The network subsequently evolves according to $s_{t+1} \sim \mathcal{P}(s_{t+1} \mid s_t, a_t)$, where the transition depends on the stochastic arrival of smart meter traffic, retransmissions, and queue evolution. The objective is to determine the optimal scheduling policy which maximises the expected cumulative discounted reward.

$$\pi^* = \arg\max_{\pi} \mathbb{E}\left[\sum_{t=0}^{\infty} \gamma^t r_t\right] \quad (9)$$

B. Joint State–Action Representation

The network state is represented by three congestion-related variables observed at the eNB scheduler, as $s = (\rho, Q, L)$, where ρ denotes the congestion level, $Q$is the retransmission backlog, and L represents the discretised offered traffic load. Uniform quantisation of these variables produces a compact state space containing 150 discrete network states. For each observed state, the scheduler jointly selects the RACH control action $a = (p_{\mathrm{ACB}}, BI, N_{\mathrm{p}})$, where p_{ACB} denotes the Access Class Barring factor, BI is the Backoff Indicator, and N_{p} represents the NPRACH preamble group configuration. Restricting these parameters to the finite sets permitted by the 3GPP NB-IoT specifications results in 120 feasible joint actions.

Consequently, the proposed controller maintains a Q-table containing $150 \times 120 = 18\,000$ state–action pairs, requiring approximately 72 KB of memory when stored using four-byte floating-point values.

C. Throughput-Oriented Reward Function and Learning Policy

The reward function directly reflects the primary objective of NB-IoT smart metering networks, namely maximising successful network access. Unlike collision-oriented reward formulations, which may encourage excessive access barring, the proposed framework employs a throughput-oriented reward based on the number of successfully completed random access procedures during each scheduling interval. This formulation naturally balances access admission, retransmissions, and NPRACH resource allocation while maintaining queue stability.

The action-value function is updated using the standard Q-learning rule

$$Q(s_t, a_t) \leftarrow Q(s_t, a_t) + \alpha \left[r_t + \gamma \max_{a'} Q(s_{t+1}, a') - Q(s_t, a_t) \right], \quad (10)$$

where α is the learning rate, γ is the discount factor, and r_t denotes the immediate throughput-oriented reward. During training, an ε -greedy exploration policy is adopted to balance exploration and exploitation until convergence.

The learning procedure follows the standard Q-learning update using ε-greedy exploration until convergence. The complete learning procedure is summarised in Algorithm 1. During online operation, the controller performs only a single Q-table lookup followed by one update for each scheduling interval, resulting in deterministic computational complexity of $O(1)$. The complete Q-table contains 18,000 state-action entries, corresponding to approximately 72 KB of memory when stored using four-byte floating-point values. This compact implementation enables practical deployment within commercial NB-IoT eNodeB schedulers without requiring neural-network inference, specialised AI accelerators, or graphics processing units.

V. Simulation Results and Discussion

A. Experimental Setup

All simulations were performed in MATLAB R2023b. Each configuration was trained for 30,000 episodes comprising 5,000 RACH scheduling intervals per episode using three independent random seeds (42, 123, and 777). Reported results correspond to the mean ± standard deviation over the final 2,000 post-convergence episodes. The principal simulation parameters are summarised in Table I.

Table I: Simulation Parameters

Parameter	Value
N_{UE} (smart meters)	50,000
T_{pre} (RACH slot)	50 ms
$N_p^{\max}$ (NPRACH subcarrier group)	48
T_{burst} (DR burst window)	10 slots
α, β (Beta shape parameters)	3, 4
D_{SLA}(study-specific delay constraint)	30 000 ms
T_{sim} (slots/episode)	5,000
Training episodes K	30,000
Seeds (reproducibility)	42, 123, 777
$\alpha_Q, \gamma, \epsilon_0$	0.25, 0.95, 1.0

The proposed framework is compared with five representative baselines:

- 3GPP Static: Fixedp_{ACB}, BI, and N_{p} without adaptive congestion control.
- LE-URC [Rxx]: Dynamic NPRACH allocation with fixed access control.
- Smart ACB-only: Aggressive fixed access barring illustrating reward-hacking behaviour.
- Single-Parameter RL: Reinforcement learning optimising only p_{ACB}.
- Dual-Parameter RL: Reinforcement learning jointly optimising p_{ACB} and BI.

All learning-based methods employ identical training episodes, exploration schedules, and convergence criteria to ensure a fair comparison.

B. Performance Comparison

Figure 3 illustrates the convergence of the proposed Q-learning framework during training, while Table II summarises the final performance comparison.

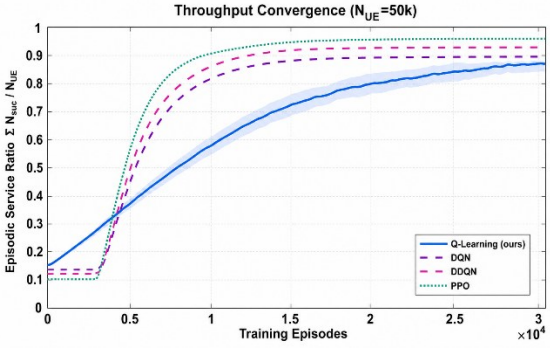


Figure 3: Throughput convergence of the proposed tabular Q-learning framework for N_{UE}=50,000 UEs.

The proposed framework consistently outperforms all baseline schemes. Conventional 3GPP Static and

LE-URC experience severe RACH congestion, whereas Smart ACB-only reduces collisions at the expense of successful network access, demonstrating reward-hacking behaviour. By jointly optimising the Access Class Barring factor, the Backoff Indicator, and the NPRACH configuration, the proposed framework achieves the highest access success probability ($P_s = 0.592$), an average access delay of 1,534 ms, and zero SLA violations.

Table II: Performance Comparison ($N_{UE} = 50\,000$, $D_{SLA} = 30000$ ms)

Method	P_c	P_s	$\overline{D}$ ms	e	SR
Q-Learning (Proposed)	0.408	0.592	1 534	1.69	0.091
± std	±0.004	±0.004	±117	—	±0.002
High-ACB Fixed	0.598	0.403	766	2.49	0.100
Smart ACB-only	0.800	0.200	21 738	5.00	0.012
Heuristic ACB+BI	0.975	0.025	99 669	19.78	0.058
LE-URC [13]	0.9998	0.0002	99 699	20.00	0.010
3GPP Static	0.9998	0.0002	99 699	20.00	0.010

Note: *P_s: per-slot; SR: preamble utilisation; SLA: $\overline{D} < 30000$ ms.*

C. Scalability and Practical Deployment

The proposed framework employs a compact Q-table containing 18,000 state-action pairs (approximately 72 KB), enabling implementation within commercial NB-IoT eNodeB platforms without specialised AI hardware. Each scheduling interval requires only one state lookup, one action selection, and one Q-value update, resulting in deterministic $O(1)$ computational complexity. Figure 4 demonstrates the scalability of the proposed framework for deployments containing 10,000, 50,000, and 100,000 smart meters. Since optimisation is performed entirely within the eNodeB scheduler by adapting the configurable 3GPP parameters p_{ACB}, BI, and N_{p}, no modification to smart meter firmware or NB-IoT user equipment is required. The framework therefore provides a practical and standards-compliant congestion-control solution for ultra-dense NB-IoT-enabled AMI deployments.

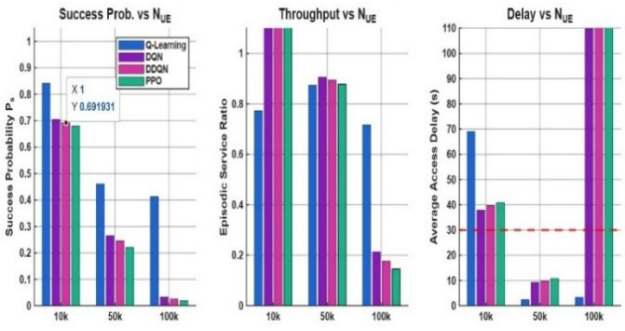


Figure 4: Scalability comparison: Q-Learning vs DRL proxies at 10 k, 50 k, and 100k Smart Meters.

VI. Conclusion

This paper presented a lightweight tabular Q-learning framework for adaptive RACH congestion control in ultra-dense NB-IoT-enabled AMI networks. The proposed framework jointly optimises the Access Class Barring (ACB) factor, the Backoff Indicator (BI), and the Narrowband Physical Random-Access Channel (NPRACH) preamble group configuration within a compact Markov Decision Process (MDP) using a throughput-oriented reward formulation. Unlike existing reinforcement learning approaches that optimise individual RACH parameters, the proposed framework enables coordinated congestion control while maintaining low computational complexity suitable for real-time implementation. Simulation results for an ultra-dense deployment comprising $N_{\mathrm{UE}} = 50{,}000$ smart meters demonstrate an access success probability of $P_{\mathrm{s}} = 0.592$, an average access delay of 1,534 ms, and zero service-level agreement (SLA) violations, outperforming conventional static and adaptive baseline schemes. Requiring only a 72 KB Q-table and deterministic $O(1)$ execution, the proposed framework provides a practical, standards-compliant, and deployment-ready solution for commercial NB-IoT radio access networks. Future work will investigate multi-cell coordination, transfer learning, and online policy adaptation using live cellular network measurements to further improve scalability and operational performance.

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